

Cyber Attack Prediction & Early Warning System Using Machine Learning

1. Introduction

The system uses Machine Learning to detect abnormal security behavior, predict potential cyber attacks, calculate risk scores, and generate early warnings for security analysts.

2. Objectives

- Detect abnormal user and network behavior.
- Predict potential cyber attacks.
- Generate risk scores and threat levels.
- Provide early-warning alerts.
- Explain high-risk predictions.
- Provide a security monitoring dashboard.

3. Scope

In Scope:

- Security data processing
- Data preprocessing
- Feature engineering
- Anomaly detection
- Attack prediction
- Risk scoring
- Alert generation
- Dashboard and historical analysis

Out of Scope:

- Unauthorized penetration testing
- Malware execution
- Vulnerability exploitation
- Offensive cyber operations

4. System Modules

1. User Authentication
2. Security Data Collection

3. Data Preprocessing
4. Feature Engineering
5. ML Prediction
6. Anomaly Detection
7. Risk Score Generation
8. Threat Classification
9. Explainable AI
10. Early Warning Alert
11. Security Dashboard

5. Functional Requirements

- Authenticate authorized users.
- Accept authorized security telemetry.
- Validate and preprocess security data.
- Extract behavioral features.
- Detect anomalous activity.
- Classify activity as Normal, Suspicious, or Potential Attack.
- Generate a risk score from 0–100.
- Generate alerts when risk exceeds the threshold.
- Provide prediction explanations.
- Display security statistics and attack trends.
- Support model evaluation and retraining.

6. Risk Classification

Risk Score	Threat Level
0–25	Low
26–50	Medium
51–75	High
76–100	Critical

7. Machine Learning

- **Random Forest / XGBoost:** Attack classification
- **Isolation Forest:** Anomaly detection
- **SHAP:** Prediction explanation
- **Future:** LSTM, Transformer, GNN

8. Technology Stack

Language: Python

ML: Scikit-learn, XGBoost/LightGBM

Deep Learning: TensorFlow/PyTorch

Backend: Flask/FastAPI

Frontend: HTML, CSS, JavaScript

Dashboard: Streamlit/Plotly

Database: MySQL/PostgreSQL

Data Processing: Pandas, NumPy

Deployment: Docker

Version Control: Git/GitHub

9. Non-Functional Requirements

- Fast prediction response
- Scalable architecture
- Reliable predictions
- Secure authentication and data storage
- User-friendly dashboard
- Modular ML pipeline
- Explainable predictions

10. Mandatory Student Tasks

Every student must complete the following:

1. **Dataset Collection**
 - Collect an authorized cybersecurity dataset.
 - Understand the features and target labels.
2. **Data Preprocessing**
 - Handle missing values and duplicates.
 - Perform data cleaning, encoding, and scaling.

3. EDA

- Analyze attack patterns.
- Create suitable visualizations.

4. Feature Engineering

- Create meaningful security behavioral features.

5. ML Model

- Implement **Random Forest/XGBoost**.
- Implement **Isolation Forest** for anomaly detection.

6. Model Evaluation

- Calculate Precision, Recall, F1-Score, and ROC-AUC.

7. Risk Score

- Generate a **0–100 risk score**.
- Assign Low/Medium/High/Critical threat levels.

8. Alert Generation

- Generate an early warning when the risk exceeds the threshold.

9. Dashboard

- Display risk score, threat level, alerts, attack trends, and indicators.

10. Documentation & Presentation

- Submit source code, dataset, results, screenshots, documentation, and final presentation.

11. Expected Output

The dashboard should display:

- Risk Score
- Threat Level
- Attack Probability
- Major Risk Indicators
- Early Warning Status
- Investigation Priority

12. Future Enhancements

- LSTM/Transformer attack-sequence prediction
- Graph Neural Network detection
- Real-time Kafka streaming

- Concept-drift detection
- Automatic model retraining
- Continuous ML monitoring

13. Expected Result

The system is expected to detect abnormal behavior, identify potential cyber attacks, recognize multi-stage attack patterns, predict attack probability, and provide explainable early warnings to security analysts.

14. Conclusion

The proposed system combines cybersecurity analytics and machine learning to move from reactive attack detection toward proactive cyber-threat prediction.